BRD. Step-1

Business Requirement Document

Requirements Analysis *Review the requirements above and prepare:* *A. Key Functional Requirements* *B. Key Non-Functional Requirements* *C. Risks & Assumptions* *List:* *• Top 5 implementation risks* *• Top 5 assumptions*

P001- Business Requirements Document

Summary	
Business Owner (SME):	Government Organisation
Analyst:	Aizhan Ismailova
Requested on	8-June-2026
Estimated Effort :	6 weeks
Status	In Progress

Project Summary

Section	Description
Executive summary/Background	A government organization and its affiliated entities have issued an RFP for a unified Visitor Management System covering pre-registration, approvals, kiosk check-in/out, QR passes, multi-entity administration, and integration with existing security infrastructure.
Problem Statement	The customer currently lacks a standardized, secure way to register, approve, and track visitors across its parent organization and affiliated entities, creating security gaps, inconsistent visitor experiences, and limited audit visibility. Manual or fragmented processes make it difficult to enforce approval policies, screen visitors, integrate with access control, and meet government security and data protection requirements. A centralized, multi-entity VMS is needed to bring governance, compliance, and operational efficiency to visitor handling.

Scope of work	**In Scope:** ● Visitor registration and pre-registration ● Approval workflows ● Visitor check-in and check-out ● Kiosk support ● QR code generation ● Multi-entity architecture ● Integration with existing systems ● Reporting and analytics ● Compliance alignment with UAE government security and data protection standards **Out of Scope:** ● Site civil works and base infrastructure (network cabling, power, counters/mounting) ● Biometric enrollment (facial recognition, fingerprint) ● replacement or upgrade of existing access control systems ● migration of historical visitor records are excluded from the baseline unless explicitly requested.
Features	The solution provides end-to-end visitor lifecycle management from pre-registration with secure QR passes, configurable multi-level approval workflows, self-service kiosk check-in/out with badge printing, to watchlist screening across a multi-entity architecture with entity-level autonomy and centralized oversight.
Limitations & Risks	Key risks are integration complexity with heterogeneous access control systems across entities, data residency constraints that may disqualify cloud-only products, and potential delays from government security accreditation.

A. Key Functional Requirements

F-1. Visitor Registration & Pre-Registration

Walk-in registration at reception/kiosk with ID capture (Emirates ID reader likely), photo capture, and visit purpose.
Pre-registration by hosts via web portal/Outlook, with invitation emails/SMS containing QR codes.
Bulk/group visitor registration (delegations, events, contractors).
Recurring visitor and long-term contractor profiles.
The system shall include watchlist/blocklist screening against internal and (potentially) government security lists.

F-2. Approval Workflows Management

Configurable multi-level approval chains (host → host manager → security), varying by visitor type, visited area/zone (building), and entity.
Auto-approval rules for low-risk visitor categories; escalation and delegation rules.
Notifications (email/SMS/app) to approvers and visitors at each stage.

F-3. Check-In / Check-Out

QR/badge shall be scanned for check-in, only pre-registered users are able to pass (see F-1).
QR/badge shall be scanned for check-out process, auto check-out rules.
Check-in and Check out processes are tracked and stored in the system.

F-4. Kiosk Support

Self-service kiosks: registration, QR scan, ID scan, photo, badge print.
Kiosks shall include Arabic/English bilingual UI (mandatory for UAE government).
Accessibility compliance and offline/degraded-mode operation.

F-5. QR Code Generation

QR-s shall be unique, time-bound, single/multi-use and issued at pre-registration
Secure (signed/encrypted) QR codes to prevent forgery/reuse.
Delivery via email, SMS, and wallet (Apple/Google) optionally.

F-6. Multi-Entity Architecture

Single platform serving the parent organization + affiliated entities, each with its own branding, workflows, approval rules, sites, and user base.
Data segregation between entities with centralized oversight/consolidated reporting for the parent.
Entity-level and global admin roles; cross-entity visitor pass scenarios.

F-7. Integration with Existing Systems

PACS (e.g., Lenel, Honeywell, HID, Genetec) to provision temporary access credentials.
Active Directory / Azure AD (SSO for hosts and admins).
Email/SMS gateways; calendar (Exchange/M365).
HR systems (host directory), possibly ERP/CRM.
UAE-specific: Emirates ID reader integration, potentially UAE PASS.

REST APIs for future integrations.

F-8. Reporting & Analytics

Real-time dashboards (visitors on site, by entity/site/zone).
Operational reports: visit history, no-shows, peak hours, host activity, dwell time.
Compliance/audit reports: full audit trail, watchlist hits, data access logs.
Scheduled/exportable reports; per-entity and consolidated views.

F-9. Security & Compliance

Role-based access control, full audit logging.
Data retention and purge policies per government regulation.
NDA/policy acknowledgement capture at check-in; consent management.

B. Key Non-Functional Requirements

NF-1. Hosting & Data Residency

In-country hosting required; data must not leave the UAE.

NF-2.Availability & Resilience

SLA ≥99.9% uptime.

NF-3.Performance & Scalability

Sub-second QR validation; peak-load support; scalable across entities and sites.

NF-4. Maintainability

Local support/SLA, safe upgrade path, admin self-service for workflow changes.

C. Risks & Assumptions

Implementation Risks

R-1. Physical Access Control System (PACS) integration complexity

Entities may run different/legacy access control systems with limited APIs; highest-effort, highest-risk item.

R-2.Data residency / hosting constraints eliminate product options

Many leading VMS are cloud-only SaaS; on-prem mandate shrinks the shortlist and adds infrastructure/security effort.

R-3.Multi-entity governance ambiguity

Unclear ownership between parent and affiliates (workflows, data, licensing) can stall design and force tenancy-model rework.

R-4. Compliance/accreditation delays

Government security review, IA audits, and pen-testing cycles can add months; remediation may be outside our control.

R-5. Hardware/kiosk supply and site readiness

Kiosks, badge printers, Emirates ID readers, turnstile integration depend on procurement lead times and site infrastructure across many locations.

Assumptions

1. Customers will permit hosting in a UAE-based environment acceptable to them (on-prem or approved sovereign cloud), confirmed before solution selection.
2. Entities will adopt a single shared platform with entity-level configuration — not separate deployments per entity.
3. Existing systems (PACS, email/SMS, HR) expose supported APIs/connectors; customer provides test environments and technical POCs.
4. Customer provides and funds site infrastructure (network, power, installation conditions); our scope = VMS software, kiosks/peripherals, integration.
5. Identity verification via Emirates ID/passport scan and QR passes; no biometrics in baseline scope unless explicitly stated.

Questions. Step-2

Clarification Questions

Step 2 – Clarification Questions *Prepare a list of:* *Functional Questions* *Technical Questions* *The objective is to reduce project uncertainty before estimating effort and cost.*

During the interview with the client I would also use a Process map, Workflow schemas for better understanding of current and future states of the client's system.

Part 1: Functional Questions

FQ1. Current Visitor Management Process

How is visitor management currently handled across the head organization and the affiliated entities? Please describe whether this is done through manual registers, an existing system, or a combination of both, and highlight the main pain points. Please also confirm whether migration of historical visitor records is expected.

FQ2. Sites and Visitor Volumes

How many entities, buildings, and entry points (receptions, gates, turnstiles) are in scope of this project? What are the average and peak daily visitor volumes at each location?

FQ3. Multi-Entity Operating Model

Should all entities operate on a single shared platform with entity-level configuration, or do certain entities require separate instances? Please clarify who will govern visitor workflows, who owns the visitor data, and how licensing costs will be allocated between the head organization and the entities.

FQ4. Visitor Categories and Approval Workflows

Which visitor categories must be supported (e.g. guests, contractors, VIP delegations, delivery personnel)? For each category, what approval workflow applies, and does the workflow change based on the sensitivity of the area being visited or differ between entities?

FQ5. Identity Verification Requirements

What level of identity verification is required during visitor registration? Please confirm whether Emirates ID or passport scanning is required, whether UAE PASS integration is desired, whether photo capture is mandatory, and whether biometric capabilities such as facial recognition are expected now or planned for the future.

Part 2: Technical Questions

TQ1. Hosting and Data Residency

Which hosting models are acceptable to your organization: on-premises, government cloud, or a UAE-based cloud service? Is there a formal data residency requirement that mandates all visitor data to remain within the UAE?

TQ2. Existing Access Control Systems

Which physical access control systems are currently deployed at each entity? Please specify the vendor, product, and version. Is the new VMS expected to issue temporary access credentials into these systems, or is visitor check-in and tracking sufficient?

TQ3. Existing IT Landscape and Integrations

Which corporate systems should the VMS integrate with? Please confirm the identity provider used for staff authentication (e.g. Active Directory, Microsoft Entra ID), the approved email and SMS gateways, the HR system that maintains the employee directory, and any SIEM or SOC platform that must receive security events.

TQ4. Security Standards and Accreditation

Which security standards and policies must the solution comply with (e.g. UAE Information Assurance Standards, ISO 27001, internal security policies)? Is a formal security accreditation and independent penetration test required before go-live, and what is the expected duration of this process?

TQ5. Hardware Scope, Site Readiness, and Timeline

How many kiosks, badge printers, and ID scanners are anticipated across all sites? Please confirm whether hardware supply and installation falls within the vendor scope or will be provided by your organization, the readiness status of the sites in terms of network and power, and whether there is a fixed go-live date the project must meet.

Solution. Step-3

Solution Assessment

> *Research 2-3 products from the market that will be a good fit:*
> *Prepare a maximum 2-page summary for each covering:*
> *Product Overview*
> *Why It Could Fit the Customer Requirements*
> *Key Advantages*
> *Potential Gaps / Risks*
> *Typical Integrations*
> *Licensing Considerations*

Solutions are presented in order of suitability, from highest to lowest fit against stated requirements.

Option 1 – HID SAFE Visitor Manager (HID Global)

Product Overview: Enterprise VMS automating the full visitor lifecycle: pre-registration, approvals, check-in/check-out, watchlist checks, badge printing, reporting with self-service kiosk and an optional Visitor Secure Enrollment Portal (visitors submit PII, scan IDs, photos, acknowledge NDAs pre-arrival; built for government customers). Part of the HID SAFE PIAM suite (centralized identity lifecycle, access workflows, governance across multi-site facilities; dedicated "SAFE for Government" edition). HID also offers EasyLobby SVM as a fully on-premises VMS is critical fallback if cloud is ruled out.

Why it fits: Covers every RFP functional requirement natively; centralized enterprise/regional policy management matches parent + affiliates; out-of-the-box connectivity to PACS, HRMS, IDMS, background-check and emergency notification systems; temporary credentials auto-provisioned into PACS for time-limited entry; government pedigree; HID hardware widely deployed in UAE government.

Key advantages: Strongest security/compliance posture; hosting flexibility (cloud or on-prem); upgrade path to full PIAM; vendor-agnostic PACS integration (Lenel, Honeywell, etc.).

Gaps/risks: Cloud product and on-prem EasyLobby are different codebases — EasyLobby is older-gen with dated UI; Arabic/RTL support must be verified; multi-entity tenancy is policy/region-based, not true multi-tenant; heavier implementation than SaaS rivals.

Typical integrations: PACS (HID, Lenel OnGuard, Honeywell, AMAG), AD/Azure AD, HRMS (SAP, Workday), Outlook/Exchange, Descartes Visual Compliance watchlists, email/SMS, REST APIs.

Licensing: Cloud = subscription (per site/user tiers); Hardware separate. Enterprise pricing via HID regional channel;

Website: https://www.hidglobal.com/products/hid-safe-visitor-manager

Option 2 – Genetec ClearID + Security Center Synergis

Product Overview: Workflow-based physical access management covering visitor invitations, contractor access automation, access reviews and compliance auditing; unified with Synergis access control (fewer integrations to maintain). Employees invite guests via self-service portal; guest receives confirmation email and checks in at kiosk, host auto-notified.

Why it fits: Multi-entity is a native strength ClearID uses a single identity to synchronize access across multiple independent sites running different Synergis instances, creating local cardholders only when first needed (minimizes PII replication). QR met natively: ClearID auto-creates QR credentials usable at parking entrances, turnstiles, gated facilities (real access credential, not just check-in token). AD-integrated; automated access reviews keep audit readiness. Genetec heavily deployed across UAE government/critical infrastructure — if customer already runs it, riskiest integration collapses to near zero.

Key advantages: VMS + access control + video unified (visitor events correlate with door events and CCTV); escort rules, area permissions, anti-passback; strong GCC presence and government references.

Gaps/risks: ClearID is cloud-hosted SaaS if fully on-prem is mandated, may be disqualified (fallback: basic on-prem Synergis visitor module); visitor-facing/kiosk UX less refined than dedicated VMS; advantage assumes Genetec PACS heterogeneous PACS erodes it; highly bespoke workflows may need workarounds.

Typical integrations: Native Synergis PACS, Security Center video, AD/Entra ID; HR feeds, email, REST APIs, Mercury/Axis hardware; partner hub incl. third-party VMS (e.g., iLobby).

Licensing: ClearID subscription (per identity/site) on top of Security Center/Synergis (perpetual + SMA or subscription). Attractive incremental cost if Security Center already owned; greenfield across all entities is a larger investment.

Website:
https://techdocs.genetec.com/r/en-US/Genetec-ClearIDTM-User-Guide/What-is-ClearID

Option 3 – iLobby VisitorOS (FacilityOS)

Product Overview: FacilityOS deployed at 6,000+ sites; VisitorOS is the enterprise visitor management module for mission-critical facilities. Interoperable modules: VisitorOS, EmergencyOS (evacuation/mustering), ContractorOS, LogisticsOS, PIAM/access control. Clients include governments, banks, airports, Fortune 500.

Why it fits: Full functional coverage with fast time-to-value: pre-registration, instant host notifications, customizable sign-in workflows, real-time monitoring, compliance/audit reporting, ID verification, e-signature NDAs, badge printing, automated screening. Multi-location dashboard with seconds-fast reports; built-in safety orientations/quizzes/agreements; EmergencyOS adds mustering. Turnkey hardware+software bundle with managed devices; reference: 90 locations rolled out in ~3 months.

Key advantages: Best visitor/kiosk UX and fastest deployment; modular pricing; bundled managed kiosk hardware lowers our hardware risk; strong contractor management (inductions, document expiry).

Gaps/risks: Cloud-only SaaS is biggest risk; UAE residency must be confirmed or likely disqualified. PACS integration shallower (QR typically check-in, not turnstile credential, without extra modules/custom work). Arabic/RTL and regional support to verify. Per-location economics: ~$199–275/month per location for VisitorOS plus $199–499/month per extra module adds up across many sites.

Typical integrations: AD/Azure AD SSO, Outlook/Google calendar, Slack/Teams, access control integrations (incl. Genetec partner integration), watchlists, REST APIs, bundled printers/scanners.

Licensing: Pure subscription per location per module, annual billing; hardware bundled/leased. Predictable OPEX; no perpetual/on-prem option.

Website: https://www.facilityos.com/

Comparison table:

#	Requirement	HID SAFE VM	Genetec ClearID	iLobby VisitorOS
F-1	**Visitor Registration & Pre-Registration** (walk-in, QR invites, bulk, recurring, watchlist)	Native	Native	Native
F-2	**Approval Workflows** (multi-level, auto-approval, notifications)	Native	Native	Native

F-3	**Check-In / Check-Out** (QR/badge scan, badge printing, auto check-out, overstay alerts, evacuation register)	Native	Native	Native
F-4	**Kiosk Support** (self-service, ID scan, photo, badge print, Arabic/English, offline mode)	Native	Moderate — kiosk UX less refined than dedicated VMS; Arabic/RTL to verify; offline to confirm	Best kiosk UX — turnkey managed hardware bundle; Arabic/RTL **to verify**; offline to confirm
F-5	**QR Code Generation** (time-bound, signed/encrypted, email/SMS/wallet delivery)	Native	Native	Native
F-6	**Multi-Entity Architecture** (single platform, per-entity branding/workflows, data segregation, consolidated reporting)	Strong	Strongest	Good
F-7	**Integration with Existing Systems** (PACS, AD/Azure AD, email/SMS, HR, Emirates ID, UAE PASS, REST APIs)	Strong,	Strong	Moderate
F-8	**Reporting & Analytics** (real-time dashboards, operational reports, compliance/audit, scheduled exports, per-entity & consolidated)	Native	Native	Native

F-9	**Security & Compliance** (RBAC, audit logging, data retention/purge, NDA capture, consent management, UAE IA/NESA/ISO 27001)	Excellent	Excellent	Good
NF-1	**Hosting & Data Residency** (in-country hosting; data must not leave UAE)	**Fully met**	**Risk**	**High risk**
NF-2	**Availability & Resilience** (SLA ≥99.9% uptime, kiosk offline mode, DR with RPO/RTO)	Strong	Moderate	Moderate
NF-3	**Performance & Scalability** (sub-second QR validation, peak-load support, scalable across entities/sites)	Strong	Strong	Moderate
NF-4	**Maintainability** (local support/SLA, safe upgrade path, admin self-service for workflow changes)	Strong	Strong	Moderate

My next steps as analyst:
I would identify the weight of each requirement criterias based on the client's interview and will create a logic for comparison score matrix to identify the best option to proceed.